

EXPLANATION:

1. 9 V Battery

Use:

- Acts as the **power source** for the circuit.
- Provides electrical energy required to light the LEDs.
- Commonly used for small experimental and breadboard circuits.

Why used here?

- Easy to connect
 - Suitable voltage for demonstration (with current limiting)
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2. Battery Snap Connector (Red & Black Wires)

Use:

- Connects the 9 V battery to the breadboard.
- **Red wire** → Positive terminal (+)
- **Black wire** → Negative terminal (-)

Why used here?

- Safe and convenient way to draw power from the battery.
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3. Breadboard

Use:

- A **solderless platform** for building and testing circuits.
- Allows quick connection and modification of components.

Internal connections:

- Vertical side rails → Power supply (+ and -)
- Middle rows → Component connections

Why used here?

- Ideal for learning, testing, and prototyping circuits.
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4.LEDs (Light Emitting Diodes)

Use:

- Convert **electrical energy into light**.
- Used as indicators, displays, and for visual output.

Color meaning (general):

- Red / Yellow → Status indicators
- Blue / White → Higher forward voltage LEDs

Why used here?

- To demonstrate how current flows and produces light.
 - Easy to visually verify circuit operation.
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5. Connecting Wires (Jumper Wires)

Use:

- Provide **electrical connections** between battery, breadboard, and LEDs.
- Complete the circuit path for current flow.

Why used here?

- Flexible and reusable
 - Essential for routing power and signals
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6. Power Rails (+ and – on Breadboard)

Use:

- Distribute voltage evenly across the breadboard.
- Makes powering multiple components easier.

Why used here?

- To supply common positive and negative connections to all LEDs.

Simple Overall Working

1. Battery supplies power.
2. Current flows through wires into the breadboard.

3. LEDs receive forward bias.
4. LEDs glow, indicating circuit completion.

CIRCUIT:

